

RAGAS VS. DEEPEVAL CHEATSHEET

AI Engineer Ready · Week 6 — RAG Optimization + Evaluation + LangSmith

Golden Rule -> same metrics, two libraries -- RAGAS for RAG-only speed, DeepEval for CI-gated full test suites

1. Why Use a Library Instead of Hand-Rolling

Faithfulness and Relevance -- the exact metrics from the last cheatsheet -- are the same idea, now as tested, versioned library code instead of a prompt only you maintain. Understanding the mechanics first (which you now do) is what lets you debug a library score that looks wrong.

2. RAGAS -- the RAG Specialist

- Started as a research paper in 2023, laser-focused on RAG.
- Core metrics: Faithfulness, Response Relevancy, Context Precision, Context Recall.
- Reference-free -- most metrics need no ground-truth labels.
- Lightweight -- a few lines to score a RAG pipeline.

```
$ from ragas import evaluate
$ from ragas.metrics import faithfulness, answer_relevancy
$ from datasets import Dataset

$ data = Dataset.from_dict({
$ "question": [...], "contexts": [...],
$ "answer": [...], "ground_truth": [...],
$ })
$ result = evaluate(data, metrics=[faithfulness, answer_relevancy])
```

3. DeepEval -- the Full Test Suite

- Built by Confident AI: "pytest for LLMs."
- 50+ metrics across RAG, agents, chatbots, and safety (bias, toxicity, hallucination).
- Custom criteria via G-Eval, plus assertions and thresholds.
- CI/CD-native -- evaluations run as gated tests, not one-off notebook cells.

```
$ from deepeval import assert_test
$ from deepeval.metrics import FaithfulnessMetric
$ from deepeval.test_case import LLMTestCase

$ test_case = LLMTestCase(
$ input=question, actual_output=answer,
$ retrieval_context=retrieved_chunks,
$ )
$ assert_test(test_case, [FaithfulnessMetric(threshold=0.7)])
```

4. RAGAS vs. DeepEval, Head to Head

	RAGAS	DeepEval
Origin	Research paper (2023), cited by OpenAI at DevDay	Built by Confident AI

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	RAGAS	DeepEval
Scope	Laser-focused: Faithfulness, Relevancy, Context Precision/Recall	50+ metrics: RAG, agents, chatbots, safety
Ground truth	Reference-free -- most metrics need no ground-truth labels	Custom criteria via G-Eval, plus assertions/thresholds
Setup	Lightweight, a few lines to score a pipeline	CI/CD-native, "pytest for LLMs"
Team features	--	Optional Confident AI cloud for collaboration and monitoring

5. Rule of Thumb

If...	Reach for...
RAG metrics are the whole job and you want minimal setup	RAGAS
You want CI-gated tests, custom criteria, or coverage beyond RAG (agents, safety)	DeepEval

Pitfall

Don't adopt a library score on faith. If a RAGAS or DeepEval number looks wrong, go back to the hand-rolled Faithfulness/Relevance mechanics to understand what the metric is actually asking the judge model.

Remember

Your hand-rolled judge and these libraries are asking the exact same question -- the libraries just give you tested, versioned, reusable code instead of a prompt buried in one notebook cell.

INTERVIEW PREP

Q: When would you pick RAGAS over DeepEval?

A: When RAG metrics are the entire evaluation job and you want something lightweight and reference-free with minimal setup.

Q: When would you pick DeepEval over RAGAS?

A: When you need CI-gated tests ("pytest for LLMs"), custom scoring criteria via G-Eval, or evaluation coverage beyond RAG, such as agents or safety metrics like bias and toxicity.

Q: What does "reference-free" mean for an evaluation metric?

A: It means the metric can be computed without a pre-written ground-truth answer -- it scores faithfulness or relevance directly from the question, retrieved context, and generated answer.